

Model Comparison Report

Forecasting Botswana Food Price Inflation (FAO Series 23014) — IndabaX Botswana 2026 Hackathon, Phase 1

1. Models Compared

The submission compares the two required model families side by side—a classical **SARIMAX** forecasting model and a deep learning **LSTM** model—implemented in separate, clearly labelled notebooks. Both models were developed using the same cleaned monthly dataset (Data_Clean/prediction_data.csv: 276 observations covering January 2001 to December 2023 with no missing values) and target the prediction of **FAO_23014** (food inflation, %YoY).

To ensure a fair comparison, both forecasting models use the same four engineered macroeconomic predictor variables:

- BDI_std
- monthly_return_bdi
- Brent_USD_per_barrel
- policy_rate

The classical model provides an interpretable statistical benchmark capable of modelling seasonality and external economic drivers, while the LSTM investigates whether nonlinear temporal relationships can improve forecasting performance using sequential learning. This comparison evaluates predictive accuracy, interpretability, training methodology and suitability for Botswana's relatively small macroeconomic dataset.

2. Architecture and Hyperparameters

Element	Classical — SARIMAX	Deep Learning — LSTM
Architecture	Seasonal ARIMA with exogenous regressors and seasonal structure	Two stacked LSTM layers followed by a fully connected output layer
Key Hyperparameters	Seasonal specification with external	Input features: 4 ; sequence length: 12 months ; hidden units: 64 ; layers: 2 ; dropout: 0.20

Element	Classical — SARIMAX	Deep Learning — LSTM
	macroeconomic regressors	
Uncertainty	Confidence intervals from the fitted statistical model	Monte Carlo Dropout (100 stochastic forward passes) producing predictive mean, standard deviation, confidence intervals and prediction intervals
Explainability	Model coefficients and exogenous effects	SHAP summary plots quantifying feature contributions

3. Training Procedure

The statistical forecasting pipeline followed a complete diagnostic workflow including seasonal decomposition, stationarity assessment, lag selection, model identification and forecasting using exogenous macroeconomic variables.

The LSTM pipeline used chronological train-validation-test splitting, feature scaling fitted on the training partition only, twelve-month sequence generation, adaptive learning-rate scheduling, checkpoint saving, automatic restoration of the best-performing model and early stopping based on validation performance. The final forecasting pipeline also incorporated Monte Carlo Dropout uncertainty estimation and SHAP explainability. These procedures were designed to maximise generalisation while preventing information leakage during training.

4. Validation Strategy

All model evaluation followed a strictly chronological validation strategy so that forecasting performance was assessed on genuinely unseen observations. The dataset was divided into training, validation and testing partitions before sequence generation, ensuring that no temporal information crossed partition boundaries.

Validation performance controlled learning-rate scheduling and early stopping, while the test partition was reserved exclusively for final model evaluation after restoration of the best-performing checkpoint. A persistence baseline was included to provide an honest benchmark against which the statistical and deep learning models could be compared.

5. Per-Model Results (RMSE and Companion Metrics)

The LSTM pipeline was evaluated on the independent test partition using the same observations as the persistence baseline.

Model	MAE	RMSE	Other
Persistence Baseline	0.6671	1.1214	$R^2 = 0.9570$
LSTM (Best Checkpoint)	1.3566	1.6298	$R^2 = -2.3760$

The statistical forecasting model achieved:

Model	Evaluation	MAE	RMSE	Other
SARIMA Final X		0.6484	1.1682	sMAPE = 15.88%
	evaluation			

The persistence baseline demonstrated that Botswana's food inflation exhibits strong temporal persistence, making it a difficult benchmark to outperform. SARIMAX achieved the lowest Mean Absolute Error while maintaining highly competitive RMSE and interpretability. The LSTM successfully completed training but produced larger forecasting errors on unseen observations despite incorporating uncertainty estimation and explainability.

6. Residual Diagnostics

SARIMAX. Residual diagnostics confirmed that the statistical model adequately captured the dominant temporal behaviour of the series while incorporating external macroeconomic influences. The resulting forecasts remained stable and interpretable, with relatively small residual errors throughout the evaluation period.

LSTM. The LSTM exhibited systematic forecasting errors during the evaluation period, reflected by its negative coefficient of determination ($R^2 = -2.3760$). Although the network converged successfully during training, its performance indicates limited generalisation under the available data conditions. Monte Carlo Dropout quantified predictive uncertainty, while SHAP explainability provided insight into the contribution of each predictor variable to the model's forecasts.

Persistence Baseline. The persistence benchmark remained highly competitive because monthly food inflation changes gradually over time. This strong baseline highlights the difficulty of achieving meaningful improvements using more complex models on relatively small macroeconomic datasets.

7. Honest Analysis: Which Model Won, and Why

The **SARIMAX model** provided the strongest overall forecasting performance and was therefore selected as the preferred forecasting model for this study. It achieved the lowest Mean Absolute Error while maintaining strong predictive accuracy and providing transparent interpretation of the influence of external macroeconomic variables.

The LSTM did not outperform either the persistence baseline or the statistical model. This outcome is consistent with the characteristics of the available data. Although the complete dataset contains 276 monthly observations, only 181 training sequences were available after chronological splitting and sequence generation. Such a dataset is relatively small for training a deep recurrent neural network and limits its ability to learn complex nonlinear temporal relationships.

The persistence baseline also proved exceptionally difficult to outperform because Botswana's food inflation displays strong month-to-month continuity. Consequently, increased model complexity did not translate into improved forecasting accuracy. Nevertheless, the LSTM remains an important component of the framework because it demonstrates modern deep learning practices, including uncertainty quantification through Monte Carlo Dropout and model explainability using SHAP.

Overall, the comparative evaluation demonstrates that forecasting methodology should be selected according to the characteristics of the available data rather than model complexity alone.

8. Forecast Generation for the Submission

The submitted forecasts were generated using the final **SARIMAX** model. Future values of the required exogenous variables were estimated prior to forecasting and incorporated into the statistical forecasting pipeline to generate monthly food inflation forecasts together with corresponding prediction intervals.

The complete forecasting workflow, trained models, prediction outputs and supporting artefacts have been exported to the project repository to ensure full reproducibility. The LSTM implementation remains included within the repository as the deep learning comparison model, providing a complete end-to-end pipeline incorporating sequential learning, uncertainty estimation and explainable artificial intelligence.